

# Author Contribution Report

Decentralized Distributed Node Health Monitoring Framework

June 26, 2026

## 1 Project Information

**Paper Title:** Decentralized Distributed Node Health Monitoring Framework  
**Course:** Distributed Systems  
**Supervisor:** Dr. Tran Huu Tam  
**Submission Date:** 27 Jun 2026

## 2 Authors

| # | Name             | Student ID |
|---|------------------|------------|
| 1 | Nguyen Ha Khai   | 10423053   |
| 2 | Nguyen Duy Khanh | 10423056   |
| 3 | Vo Minh Duy      | 10423028   |
| 4 | Ninh Hoang Quan  | 10423097   |
| 5 | To Huynh Phuc    | 10423093   |

## 3 Contribution Matrix (CRediT)

| Contribution Role             | NHK | NDK | VMD | NHQ | THP |
|-------------------------------|-----|-----|-----|-----|-----|
| Conceptualization             | P   | S   | S   | P   | S   |
| Methodology                   | S   | P   | S   | S   | P   |
| Software Development          | P   | P   | S   | P   | S   |
| Validation                    | S   | S   | –   | S   | P   |
| Formal Analysis               | P   | P   | P   | –   | –   |
| Data Collection / Experiments | –   | –   | P   | P   | P   |
| Visualization                 | S   | S   | P   | S   | P   |
| Writing                       | P   | P   | P   | P   | P   |
| Project Administration        | P   | –   | –   | P   | S   |
| Literature Review             | –   | P   | P   | –   | –   |

Symbols are defined as follows:

- P : Primary Contributor
- S : Supporting Contributor
- – : No Contribution

## 4 Detailed Contributions

### Author 1: Nguyen Ha Khai

#### Primary Roles

- Co-leader with Ninh Hoang Quan
- Algorithm analyst
- Manuscript Editor

#### Specific Contributions

- Implement FINDING NEMO & CHUALANH
- Write the FINDING NEMO component in implementation details and proposed approach section
- Write the system design part in related work section

### Author 2: Nguyen Duy Khanh

#### Primary Roles

- Software engineer
- Algorithm designer

#### Specific Contributions

- Implement hybrid SWIMPHI
- Design CHUALANH algorithm
- Write the hybrid SWIMPHI component in implementation details and proposed approach section
- Write the failure detection part in related work section

### Author 3: Vo Minh Duy

#### Primary Roles

- Frontend developer
- Literature reviewer

#### Specific Contributions

- Design the monitoring dashboard for data collection and system performance evaluation
- Review 90% citations in the research paper
- Research the architecture of the system
- Write the abstract, introduction, problem statement, and complete architectural diagram

## Author 4: Ninh Hoang Quan

### Primary Roles

- Backend developer
- Leader

### Specific Contributions

- Implement the backbone or foundation of nodes in the system
- Implement the backend server monitoring application
- Contribute to the implementation of CHUALANH algorithm
- Write the CHUALANH algorithm in implementation details and proposed approach section
- Write Server Application part in the implementation details

## Author 5: To Huynh Phuc

### Primary Roles

- Manual tester
- Code reviewer
- Algorithm designer

### Specific Contributions

- Design FINDING NEMO and hybrid SWIMPHI algorithm
- Write the evaluation; conclusion and future work sections
- Refactor the source code and ensure cohesive interactions between system components

## 5 Estimated Contribution Percentages

| Author   | Contribution (%) |
|----------|------------------|
| Author 1 | 20               |
| Author 2 | 20               |
| Author 3 | 20               |
| Author 4 | 20               |
| Author 5 | 20               |
| Total    | 100%             |

## 6 Declaration

We certify that the contribution information provided in this report accurately reflects each author's participation in the project and manuscript preparation.

**Author Signature**

**Date**

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